

HAMA, Syrian Arab Republic

WMO: 400300

Lat: 35.117N

Lon: 36.750E

Elev: 994

StdP: 14.18

Time Zone: 2.00 (SYR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.9	21.4	16.6	42.2	24.8	19.6	43.1	18.8	54.0	13.7	54.7	0.3	110	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.2	103.2	70.1	100.3	69.5	97.7	68.7	73.7	93.8	72.3	92.6	71.3	91.4	3.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	104.9	83.5	65.8	98.8	81.9	64.6	94.6	80.7	38.0	93.2	36.7	93.4	35.8	91.4	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.6	13.3	10.4	DB	26.5	108.4	4.5	3.2	23.3	110.7	20.6	112.6	18.1	114.3	14.8	116.6	(4)
(5)			WB	25.0	77.5	4.5	2.7	21.8	79.4	19.2	81.0	16.7	82.5	13.4	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.3	46.2	49.4	55.3	63.0	72.4	80.4	84.8	85.4	79.5	70.9	57.9	48.9	(6)	
	DBStd	14.82	4.76	5.24	5.33	6.20	5.50	4.31	3.33	3.46	3.91	5.76	6.05	5.14	(7)	
	HDD50	311	134	67	15	2	0	0	0	0	0	0	11	82	(8)	
	HDD65	2175	582	438	305	107	6	0	0	0	0	16	222	499	(9)	
	CDD50	6248	17	49	178	391	695	912	1080	1097	884	649	248	48	(10)	
	CDD65	2636	0	0	3	46	236	462	615	632	434	200	8	0	(11)	
	CDH74	31321	0	1	45	570	2632	5624	8024	8192	4550	1623	60	0	(12)	
	CDH80	16668	0	0	9	174	1178	3087	4664	4760	2223	569	4	0	(13)	
(14) Wind	WSAvg	2.4	1.2	2.1	2.6	2.9	3.1	3.9	4.3	3.1	2.5	1.3	1.0	0.8	(14)	
(15) Precipitation	PrecAvg	16.2	3.4	2.8	2.0	1.0	0.6	0.1	0.0	0.1	0.5	1.0	1.9	3.0	(15)	
	PrecMax	26.0	7.9	8.4	4.8	2.1	3.5	1.3	0.1	1.3	4.1	3.5	4.4	5.7	(16)	
	PrecMin	7.6	0.5	0.4	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	(17)	
	PrecStd	4.3	1.7	1.7	1.1	0.6	0.8	0.3	0.0	0.2	0.9	0.8	1.2	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.6	69.9	80.1	90.6	99.0	105.2	108.0	107.3	100.3	93.5	79.4	65.7	(19)	
		MCWB	53.1	56.1	58.5	65.0	66.0	69.0	70.5	71.5	69.4	65.4	60.3	54.5	(20)	
	2%	DB	59.9	65.0	73.4	84.6	94.9	100.3	103.0	102.8	96.2	89.5	75.3	62.2	(21)	
		MCWB	51.8	53.5	56.2	61.9	65.2	68.2	69.9	71.2	67.7	65.3	58.8	53.1	(22)	
	5%	DB	57.0	61.9	69.7	80.2	90.9	97.0	99.9	100.0	93.8	86.2	72.0	59.3	(23)	
		MCWB	50.4	52.1	55.3	60.3	63.7	66.7	69.4	70.2	67.3	64.5	57.5	52.1	(24)	
	10%	DB	54.2	58.9	66.4	76.2	87.1	93.9	97.0	97.2	91.2	82.9	68.9	56.9	(25)	
		MCWB	49.1	50.9	54.2	58.9	62.9	66.2	68.8	70.1	67.2	63.3	56.4	51.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	57.4	61.1	67.8	68.9	72.1	74.7	76.1	72.8	69.2	62.6	57.4	(27)	
		MCDB	61.1	68.0	74.2	85.2	89.1	95.5	96.8	96.1	90.8	85.1	74.6	62.0	(28)	
	2%	WB	52.9	54.8	58.3	63.9	66.8	70.1	72.9	74.4	71.0	67.2	60.8	54.9	(29)	
		MCDB	58.4	63.0	68.9	79.3	87.0	93.3	94.5	94.0	89.0	83.0	71.5	59.6	(30)	
	5%	WB	51.2	52.9	56.3	61.7	65.4	68.8	71.7	73.0	69.9	66.0	59.3	53.6	(31)	
		MCDB	55.9	60.2	67.3	76.9	85.7	91.2	93.5	92.5	87.5	81.4	68.9	57.8	(32)	
	10%	WB	49.6	51.3	54.7	59.9	64.1	67.6	70.7	71.9	68.9	64.9	57.7	52.1	(33)	
		MCDB	53.6	58.0	64.9	73.9	83.9	89.3	91.7	91.5	85.6	79.1	66.3	56.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	16.0	19.0	21.9	25.1	24.9	23.8	24.2	23.1	21.8	19.7	14.6	(35)	
		MCDBR	18.7	21.0	25.2	29.1	31.4	30.4	29.1	28.3	27.4	26.2	24.6	19.1	(36)	
		MCWBR	11.7	11.7	12.1	12.7	11.0	10.0	8.8	7.7	7.9	10.1	11.9	11.6	(37)	
	5% WB	MCDBR	15.7	19.0	22.4	26.3	26.9	26.1	24.0	24.9	23.2	22.4	19.8	14.7	(38)	
		MCWBR	10.3	11.2	11.6	12.5	10.3	9.4	8.2	7.6	6.9	9.0	10.2	9.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.368	0.395	0.430	0.484	0.473	0.420	0.425	0.437	0.420	0.438	0.385	0.371	(40)	
	taud		2.276	2.182	2.079	1.943	2.016	2.200	2.202	2.157	2.205	2.125	2.243	2.266	(41)	
	Ebn at Noon		259	265	265	256	260	274	272	266	265	246	249	248	(42)	
	Edh at Noon		33	41	49	59	56	46	46	47	43	43	34	32	(43)	
(44) All-Sky Solar Radiation	RadAvg		745	1024	1484	1865	2246	2549	2496	2245	1870	1361	965	698	(44)	
	RadStd		68	117	91	119	123	52	43	49	66	69	77	78	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		+0.54	N/A	-2.23	N/A	N/A	N/A	N/A	N/A	+311	+198	

Nomenclature: See separate page